

Quantum Mechanics Notes

1. Introduction

Quantum Mechanics studies matter and energy at atomic scales. Unlike classical physics, it is probabilistic in nature.

2. Wave-Particle Duality

Particles like electrons behave both as waves and particles. De Broglie proposed $\lambda = h/p$.

3. Uncertainty Principle

Heisenberg stated that position and momentum cannot be precisely known simultaneously.

4. Schrödinger Equation

Describes evolution of quantum states: $i\hbar \partial\psi/\partial t = H\psi$

5. Wave Function

ψ gives probability density $|\psi|^2$.

6. Quantization

Energy levels are discrete, e.g., $E_n = -13.6/n^2$ eV.

7. Superposition

Systems can exist in multiple states until measured.

8. Entanglement

Particles share linked states regardless of distance.

9. Measurement Problem

Observation collapses wavefunction.

10. Quantum Tunneling

Particles can pass barriers without enough classical energy.

11. Operators

Physical quantities are operators, e.g., momentum = $-i\hbar d/dx$.

12. Hamiltonian

Represents total energy of system.

13. Eigenvalues

Observable values obtained from operators.

14. Pauli Exclusion

No two fermions share same state.

15. Quantum Numbers

n, l, m, s define electron states.

16. Spin

Intrinsic angular momentum of particles.

17. Applications

Used in semiconductors, lasers, MRI, quantum computing.